Udhayakumar M

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I'm a B.Tech Computer Science graduate with hands-on experience in **AI, Data Analytics, and Full-Stack Development**. I work with **Python, SQL, Pandas, NumPy, Excel, Power BI, Tableau, Node.js, React, and LLMs** to build intelligent solutions and perform data-driven analysis. My expertise includes **machine learning, NLP, computer vision, and full-stack web development**. I am passionate about leveraging **data and AI** to solve real-world problems and continuously enhance my technical skills.

EDUCATION

B.Tech(CSE), Karunya Institute of Technology and Science, *CGPA: 7.16* 10/2020 – 05/2024

HSC, Mount Zion Matriculation Higher Secondary School, *68.5%* 06/2019 – 03/2020

SSLC, Mount Zion Matriculation Higher Secondary School, *78.8%* 06/2017 – 03/2018

SKILLS

Programming & Scripting: Python, C, JavaScript, SQL, Node.js, React

Analytics & Visualization: Power BI, Tableau, Excel

AI & Machine Learning: Machine Learning, Data Science, NLP, Computer Vision, LLMs (LLaMA 4)

Web & FullStack Development: HTML, CSS, JavaScript, Flask, Node.js, Express, Socket.io, MongoDB

Tools and Libraries: Groq SDK, VS Code, Postman, Git/GitHub

EXPERIENCE

Data Analyst Intern, *Besant Technologies, Chennai* 07/2025 – 12/2025

- Used Python with Pandas and NumPy to clean, process, and transform datasets for analysis.
- Wrote SQL queries with joins and aggregations to retrieve data for reporting and analysis.
- Verified and validated data to ensure accurate reporting for tasks like sales analysis and customer segmentation.

Data Science Intern, *Exposys Data Labs, Bangalore* 12/2023 – 01/2024

- Implemented customer segmentation using K-Means clustering with Python for data preprocessing and analysis.
- Performed feature extraction and data normalization to prepare structured datasets for clustering.
- Analyzed and visualized clustering results to validate and understand customer groups.

Industrial Trainee, *Intel Corporation, Bangalore* 05/2023 – 07/2023

- Implemented a regression-based mobile price prediction solution by developing Python logic for data preprocessing, model training, and evaluation.
- Organized model code into modular, reusable components to improve maintainability and simplify testing.
- Worked with team members to support integration of the trained model into internal forecasting and analysis workflows.

PROJECTS

Sales Analytics Reporting System, *Power BI, Tableau, Python, SQL, Excel*

- Developed a reporting system using SQL and backend data processing for sales and customer analysis.
- Created KPIs and automated reporting workflows for efficient data updates.
- Improved report performance and accuracy through data cleaning and validation.

AI LLM Chatbot, *Node.js, Express, Groq LLaMA 4, HTML/CSS/JS*

- Developed an AI chatbot using LLaMA 4 to answer user queries in plain text.
- Built a responsive chat interface with a scrollable conversation area.
- Created a Node.js backend with Express to handle AI requests and return real-time responses.

Weather App, *Python, Flask, OpenWeatherMap API, HTML, CSS, JavaScript, Bootstrap, Git, GitHub.*

- Built a real-time Weather App using Flask and OpenWeatherMap API to show live weather for any city.
- Created a responsive interface using Bootstrap to display temperature, weather description, and icons.
- Connected the Python backend with HTML/CSS and JavaScript fetch() to show weather results instantly.

Tic Tac Toe -- Full Stack, *React, Node.js, Express, Socket.io, MongoDB*

- Built a real-time multiplayer Tic Tac Toe game with full-stack architecture.
- Implemented live game updates using Socket.io and managed game state with Node.js backend.
- Designed a responsive React frontend and persistent game data storage in MongoDB.

Hand Gesture Recognition & Sentence Generation System, *Python, TensorFlow, OpenCV, LSTM*

- Developed a real-time hand gesture recognition system using deep learning with about 92% accuracy.
- Processed video input and predicted gestures using an efficient model pipeline.
- Converted recognized gestures into sentences using an LSTM-based text generation model.

CERTIFICATES

Google Data Analytics Professional Certificate — Coursera • Artificial Intelligence Fundamentals — IBM • Data Visualization — Kaggle • Data Mining — NPTEL • Introduction to IoT & Cybersecurity — Cisco

PUBLICATIONS

Investigation of Hand Gesture Recognition and Script Generation Models

Explored **real-time gesture recognition** and **contextual sentence generation using deep learning** for assistive communication technologies.